

The local Bloch radius as an entanglement readout: what the qg symmetry filter restores and what it does not

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Abstract

Each qubit's local qg values (qg_X, qg_Y, qg_Z) are a point in the Bloch ball. Its squared distance from the centre, $r^2 = qg_X^2 + qg_Y^2 + qg_Z^2 = 2 \text{Tr} \rho_q^2 - 1$, is the radius of the sphere on which the qubit lives (area $4\pi r^2$). For a pure register the deficit $1 - r^2$ is the known one-tangle of the qubit with the rest; with noise it mixes entanglement and impurity. We show that circuits which conserve the Hamming weight remove this ambiguity under one noise model. In a state of fixed weight $qg_X = qg_Y = 0$ on every qubit, so $r = |qg_Z|$ and the deficit is read from Z-basis shots alone; under equal amplitude damping the qg symmetry filter returns the noiseless state exactly, so the filtered deficit is the noiseless one-tangle. In a pre-registered simulation (90 configurations) the filtered deficit equals it to 3×10^{-15} , and on product states it is exactly zero, while the unfiltered readout reports a false deficit of 0.60–0.95. On five device noise models (three IBM backends, two IonQ) the filter cuts the deficit error of trained circuits 3.2–4.0 times, but it removes only 15–33% of the false deficit of a product state: errors that move an excitation inside the sector survive the filter, and near a pole the deficit amplifies them as $4\varepsilon(1 - \varepsilon)$. The radius also separates the polar angle in radians from the length of the vector, $\theta = \arccos(qg_Z/r)$: this angle is exact under depolarizing noise, where $\arccos qg_Z$ errs by up to 0.7 rad, and it is 5–9 times more accurate on the device noise models once the noise gates are actually executed. Five echo circuits measure the in-sector errors, and inverting their transfer matrix removes 38–47% of what the filter leaves, so that filter plus calibration is 6–7 times more accurate than the raw readout. Three of twenty-seven predictions failed and are reported. Everything is reproduced by the open-source library `qang` (version 0.6.11, `pip install qang`).

1 The question

The `qang` project describes a qubit by measurement-level quantities, the polar bias $qg_Z = \langle \sigma_z \rangle$ and its companions [1]. A natural addition, proposed in this project, is the sphere on which each qubit lives: its radius r , or equivalently its area $4\pi r^2$. A qubit on the surface is pure; a qubit inside the ball is mixed, either because it is entangled with other qubits or because noise made it so. The radius is old physics. For a pure register $1 - r^2$ is the one-tangle [2], for two qubits the squared concurrence [3], and its mean over the qubits is the Meyer–Wallach global entanglement [4, 5]. The question of this note is practical: when can a measured radius be read as entanglement on noisy hardware, and does the qg symmetry filter [6] help?

2 Definitions

For qubit q , $r_q^2 = qg_X^2 + qg_Y^2 + qg_Z^2 = 2 \text{Tr} \rho_q^2 - 1$, the area is $4\pi r_q^2$ and the surface deficit $4\pi(1 - r_q^2)$. We work with $D_q = 1 - r_q^2$, which is linear in the purity. If the register is pure, $D_q = 4 \det \rho_q$, the one-tangle; the mean $\bar{D}$ is the Meyer–Wallach Q . If the register is mixed, D_q adds impurity to entanglement and cannot separate the two.

Reading r_q for every qubit needs three settings for the whole register (all X, all Y, all Z). From N shots of one axis, $\hat{q}^2$ is biased upwards by $(1 - q^2)/N$, and

$$\hat{q}^2 = \frac{N\hat{q}^2 - 1}{N - 1} \quad (1)$$

is exactly unbiased (`qang.statistics.qg2_unbiased`).

In qg units the gates and algorithms have a simple radius reading [7]. One-qubit gates rotate the sphere and keep r , also for mixed qubits; SWAP exchanges radii; CNOT, CZ, iSWAP, Toffoli and Fredkin can move a product input to the centre and back. Product outputs (Bernstein–Vazirani, the QFT of a basis state) stay on the surface; Simon’s and Shor’s counting registers sit near the centre, where the answer lives in correlations; Grover’s radius dips to $r^2 = 0.66$ ($n = 5$) and returns to 1 at the optimum; a Hadamard test of U on $|\psi\rangle$ leaves the control with $r^2 = |\langle\psi|U|\psi\rangle|^2$, which is 1 only for an eigenstate.

3 Fixed weight and the filter

Fixed weight. A local X_q or Y_q changes the Hamming weight by one. If ρ has no coherence between different weights, as any state prepared and evolved inside one weight sector, then $\langle X_q \rangle = \langle Y_q \rangle = 0$ and

$$r_q = |qg_Z^{(q)}|, \quad D_q = 1 - (qg_Z^{(q)})^2 = 4p_q(1 - p_q), \quad (2)$$

with p_q the excitation probability of qubit q . The radius is read from the Z-basis shots that are already taken for the readout, and the qg filter can act on it. Amplitude damping and dephasing keep the state block-diagonal in the weight, so Eq. (2) also holds for the noisy, unfiltered state; only its meaning changes.

The filter under equal T1. On a weight- k sector the no-decay branch of equal amplitude damping on every qubit is $(1 - \gamma)^{k/2}$ times the identity. Keeping only weight- k shots therefore returns the noiseless state exactly, with kept fraction $(1 - \gamma)^{kd}$ after d layers [8]. The filtered D_q is then the one-tangle of the noiseless pure state. Unequal T1 and dephasing break this.

4 Simulation under T1

Random weight-conserving circuits of reconfigurable beam splitters (`qang.qml.WeightQNN`) on 4 and 6 qubits, weight 1 to $n/2$, depth 12–48, $\gamma = 0.02$ per qubit per layer, three noise models, 1000 and 10000 shots, 8 circuits $\times$ 20 repetitions; truth is the noiseless deficit. Six predictions were committed before the run (RESEARCH_NOTES §98).

noise	exact error, with qang	without qang	filter MSE lower	MSE ratio
equal T1	$< 3 \times 10^{-15}$	0.061–0.370	30 of 30	7.6–1005
unequal T1	0.013–0.045	0.070–0.342	30 of 30	6.8–55
dephasing 0.01	0.031–0.307	0.080–0.352	24 of 30	0.8–4.1

Table 1: Error of the radius deficit, simulation. Exact: mean absolute error of the infinite-shot deficit. MSE: from shots, with the estimator of Eq. (1); ratio of the unfiltered to the filtered MSE.

Under equal T1 the filtered deficit is the one-tangle (Fig. 1a). On product states (basis inputs, idle circuits) it is exactly 0 under all three noise models; the unfiltered readout reports 0.60–0.95 on every excited qubit: decay looks like entanglement. The unbiased estimator showed no bias

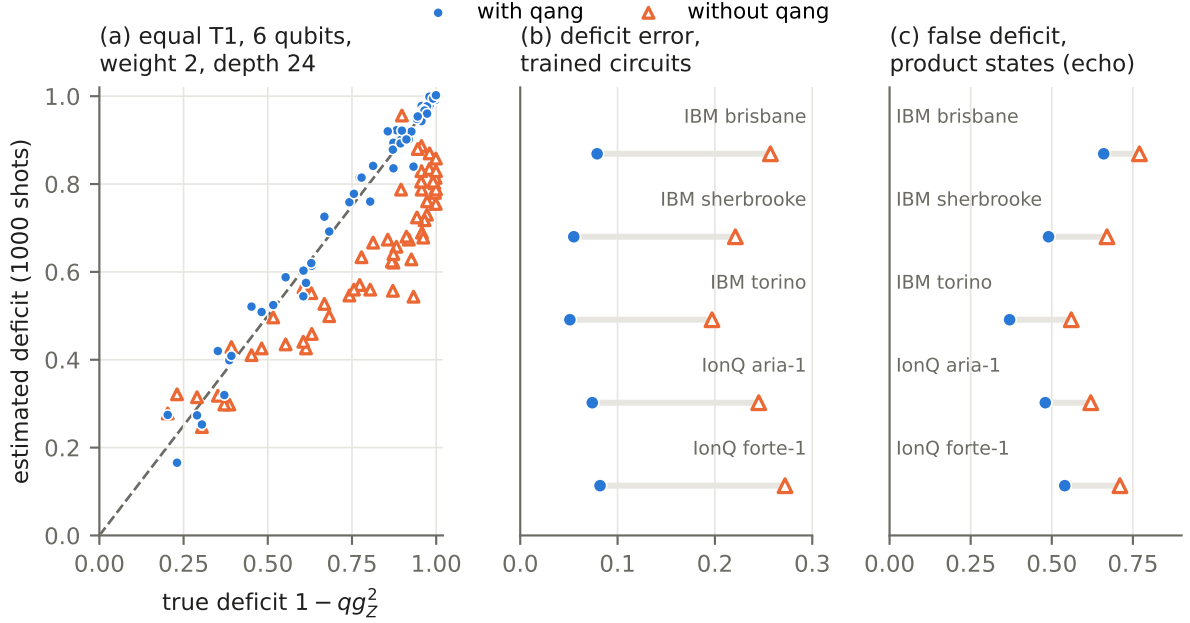


Figure 1: (a) Estimated against true deficit, equal T1, 10 random circuits on 6 qubits, 1000 shots. (b) Mean absolute error of the deficit on 30 trained circuits, five device noise models. (c) False deficit of the excited qubit of a product state after an echo circuit (truth 0).

in 30 of 30 configurations. Five predictions passed. One failed: we expected the unfiltered error to be smaller at half filling, by a first-order cancellation near $p = 1/2$; it was larger (0.177 against 0.074 for $n = 4$, depth 12), because occupations spread across qubits and more excitations decay.

5 Device noise models

A device adds gate errors, readout errors and time that a compiled two-qubit gate spends outside the weight sector, where a decay can be rotated back in. We used the five-qubit weight-1 classifier of Ref. [8] compiled to Qiskit (19 RBS gates, 38–62 native two-qubit gates), on three IBM fake backends (Aer) and the IonQ cloud simulator with the aria-1 and forte-1 noise models, 1000 shots. Trained circuits: 30 iris test inputs, true mean deficit 0.37. Echo circuits: an excited basis state, the 15 trained gates and their inverse, truth 0. Four predictions were committed before any noisy run (RESEARCH_NOTES §100). The IonQ circuits were sent in IonQ’s standard gate set; as found in Sec. 7, its compiler may simplify circuits that contain identities. A rerun in the native gate set, with its own predictions (§104), gave the same picture: 3.2–3.3 times less deficit error with the filter on trained circuits, and a false deficit on product states of 0.74 (aria-1) and 0.71 (forte-1) without the filter, of which the filter removed 17%; the standard-gate-set echo rows had understated the noise on aria-1 only.

On trained circuits the filtered deficit errs by 0.05–0.08, 3.2–4.0 times less than the unfiltered one (Fig. 1b); three predictions passed on all five backends. The fourth failed on all five: we predicted that the filter would remove more than half of the false deficit of a product state, and it removed 15–33% (Fig. 1c). The filter discards shots that left the sector, but errors that move the excitation inside the sector pass it; here they affect 10–21% of the kept shots on the excited qubit. Near a pole the deficit amplifies them: a fraction ε of misplaced shots reads as

$$D = 4\varepsilon(1 - \varepsilon), \quad (3)$$

so 10% already reads as 0.36. Where the true deficit is large the same errors change it little, which is why the trained circuits fare well.

backend	deficit error, trained with / without qang	kept shots	false deficit, echo with / without qang	two-qubit gates trained / echo
IBM brisbane	0.079 / 0.257	0.69	0.66 / 0.77	62 / 108
IBM sherbrooke	0.055 / 0.221	0.72	0.49 / 0.67	62 / 108
IBM torino	0.051 / 0.197	0.74	0.37 / 0.56	62 / 108
IonQ aria-1	0.074 / 0.245	0.74	0.48 / 0.62	38 / 60
IonQ forte-1	0.082 / 0.272	0.70	0.54 / 0.71	38 / 60

Table 2: The radius deficit on device noise models (simulated, not hardware).

6 Calibrating the in-sector errors

The errors that survive the filter can be measured. An echo circuit that prepares the sector state $|e_j\rangle$ and runs the circuit and its inverse should return $|e_j\rangle$; its filtered distribution over the sector states is column j of a transfer matrix M , with $M_{ij} = P(\text{measured } i \mid \text{prepared } j)$. The forward circuit has about half the echo’s gates, so we model its in-sector error by $M^{1/2}$ and invert it on the filtered distribution of each trained circuit (non-negative least squares on the simplex), as readout mitigation does for bit flips (`qang.sectors.echo_transfer_matrix`, `unmix_sector`). Four predictions were committed before the run (RESEARCH_NOTES §105); five echo circuits at 4000 shots calibrate each backend.

backend	without qang	qang	qang + echo ($M^{1/2}$)	qang + echo (M)
IBM brisbane	0.264	0.082	0.043	0.053
IBM sherbrooke	0.231	0.059	0.032	0.038
IBM torino	0.197	0.050	0.031	0.034
IonQ aria-1 (native)	0.301	0.090	0.049	0.050
IonQ forte-1 (native)	0.285	0.089	0.047	0.039

Table 3: Mean absolute error of the radius deficit on the 30 trained circuits, with the echo calibration.

All four predictions passed. The calibration removes 38–47% of the error the filter leaves, and filter plus echo is 6–7 times more accurate than the raw readout on every backend (Table 3). The square root beats the full matrix on average, but not on forte-1, where the forward circuit’s in-sector error is closer to the echo’s own. The diagonal of M (0.77–0.92) measures what the filter cannot clean.

7 The angle in radians, separated from the radius

The polar bias mixes direction and length, $qg_Z = r \cos \theta$, so the usual angle readout $\theta = \arccos qg_Z$ drifts towards $\pi/2$ whenever noise shortens the vector. The radius separates them:

$$\theta = \text{atan2}(\sqrt{qg_X^2 + qg_Y^2}, qg_Z) = \arccos(qg_Z/r). \quad (4)$$

Depolarizing noise leaves Eq. (4) unchanged; dephasing shrinks only qg_X, qg_Y and pulls it towards the poles, while $\arccos qg_Z$ is then exact; amplitude damping moves both. We compared the two readouts at equal total shots, 3000 per angle: with qang, Eq. (4) from 1000 shots in each of the X, Y, Z bases (equatorial part from unbiased squares); without qang, $\arccos qg_Z$ from 3000 Z shots (RESEARCH_NOTES §102, six predictions).

Under depolarizing noise Eq. (4) is exact while $\arccos qg_Z$ errs up to 0.70 rad near the poles; with shots its RMSE is 5.8 times lower at $p = 0.2$. Under dephasing it loses, as predicted, and without noise the three bases cost a factor 1.6 in RMSE. Under amplitude damping it errs

noise (simulation, 31 angles)	with qang	without qang
depolarizing 0.1 / 0.2 / 0.3	0 / 0 / 0	0.122 / 0.213 / 0.294
dephasing, coherence 0.9 / 0.8 / 0.7	0.035 / 0.073 / 0.116	0 / 0 / 0
amplitude damping 0.1 / 0.2 / 0.3	0.072 / 0.159 / 0.270	0.150 / 0.270 / 0.382
device noise model (11 angles, 16 two-qubit gates)		
IBM brisbane / sherbrooke / torino	0.053 / 0.029 / 0.041	0.267 / 0.207 / 0.374
IonQ aria-1 / forte-1, native gates	0.028 / 0.026	0.249 / 0.226

Table 4: Mean absolute error of the polar angle (rad). Simulation: exact values; device noise models: 3000 shots per angle.

29–52% less. On the IBM noise models the vector shrinks almost uniformly and Eq. (4) is 5–9 times more accurate. On the IonQ simulator the first run failed: the noise block (identity pairs of CX) was removed by IonQ’s compiler, so the vector did not shrink and the extra Z shots won. A follow-up with its own predictions (§103) confirmed this: in IonQ’s standard gate set the block changed nothing, and sent as 16 native MS gates, which IonQ runs as given, it shrank the vector and Eq. (4) was 8.7–9.0 times more accurate. The rule is simple: if the measured radius is close to 1, read $\arccos qq_Z$ with all shots in Z; if it is clearly below 1 and the noise is not pure dephasing, use Eq. (4). For circuits that contain identities, such as the echo circuits above, IonQ requires the native gate set.

8 Discussion

The radius is a measure of entanglement only when the register is pure. Weight conservation and the qq filter make it so exactly under one noise model, equal T1, and they do it from the Z-basis shots of the readout, with a shot cost known in advance. On device noise the filtered radius remains a good estimate of a large deficit, but it is not a reliable test of “no entanglement”: in-sector errors, invisible to the filter, make product states look entangled. A small deficit therefore needs a calibration; the echo calibration of Sec. 6 removes nearly half of the remaining error.

What is new and what is not. The radius, its relation to purity, the one-tangle and the Meyer–Wallach measure are known [2–5], and so is symmetry verification [9, 10]. What we add is the combination: that in fixed-weight circuits the radius is $|qq_Z|$ and is read from Z-basis shots, that the filter makes it an entanglement measure under equal T1, and measurements of how much of that survives unequal T1, dephasing and device noise models, with every prediction committed before its run, and the angle in radians that the radius frees from the noise.

Limitations. All noise is simulated, including the device noise models. The circuits are small (4–6 qubits). The echo circuits measure the false deficit of one family of product states. No quantum advantage is claimed.

Code and data availability

Library qang 0.6.11 (MIT, pip install qang): `qang.formulation (radius_profile, radius_deficit, sphere_area, meyer_wallach, weight_sector_radius, gate_radius_class, algorithm_radii, direction_from_qg), qang.sectors (echo_transfer_matrix, unmix_sector), qang.statistics (qq2_unbiased, radius2_estimate, direction_estimate)`. Studies: `examples/qg_radius_witness_qg.py` (§98) and `examples/qg_radius_hardware_qg.py` (§100), `examples/qg_direction_radians_qg.py` (§102) and `examples/qg_direction_ionq_native_qg.py` (§103), with predictions in their

headers and tests in `tests/`; figure: `manuscript/radius/make_figure.py`. Repository: <https://github.com/Viny2030/qang>.

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